

Supplementary Material

This file contains the following appendices:

1. **Appendix A** contains material relevant to RQ-1. Specifically, it includes:
 - Table A.1, which presents the types of anomalies in thirteen publicly available datasets.
 - Table A.2 - this table presents the distribution of normal and anomalous traffic across widely used cybersecurity datasets, highlighting the most dominant anomaly categories in each dataset and illustrating the significant class imbalance that characterizes many benchmark datasets used for anomaly detection research.
2. **Appendix B** contains material relevant to RQ-2. Specifically, it includes:
 - Table B.1 - This table provides a summary of the Value-based approaches of Reinforcement Learning used in cybersecurity.
 - Table B.2 - This table provides a summary of the Policy Gradient-based approaches of Reinforcement Learning used in cybersecurity.
 - Table B.3 - This table provides a summary of the Hybrid Actor-Critic-based approaches of Reinforcement Learning used in cybersecurity.
 - Table B.4 - This table provides a summary of the state-of-the-art studies based on RL algorithms for cybersecurity.
3. **Appendix C** contains material relevant to RQ-3. Specifically, it includes:
 - Table C.1 - this table provides a summary of GAN-based generative AI approaches used in cybersecurity, highlighting their output modalities, training paradigms, strengths, and limitations for tasks such as intrusion detection, phishing generation, and traffic synthesis.
 - Table C.2 - this table presents VAE-based generative AI approaches applied in cybersecurity, emphasizing their probabilistic training mechanisms and their usefulness in anomaly detection and representation learning.

- Table C.3 - this table summarizes transformer and large language model (LLM)-based generative AI techniques used in cybersecurity, particularly for language-driven tasks such as log analysis, alert classification, and phishing detection.
- Table C.4 - this table outlines hybrid and emerging generative AI approaches for cybersecurity, including combinations of GANs, VAEs, transformers, and diffusion models that aim to improve data generation quality, contextual modeling, and adaptive threat simulation.
- Table C.5 - this table provides a summary of the state-of-the-art studies based on Gen AI for cybersecurity.

Appendix A. Tables Relevant to RQ-1

Table A.1: Publicly available datasets for anomaly detection.

Dataset	Types of anomalies
UNSW-NB15	9 cat.: Fuzzers, Analysis, Backdoor, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms
SWaT	Normal and abnormal
MIMIC-III	Policy violation
CIC-IDS2017	15 cat.: DoS, PortScan, Web (XSS, SQLi), Heartbleed, etc.
N-BaIoT	2 cat.: Mirai, BASHLITE
Bot-IoT	4 cat.: DDoS, DoS, Reconnaissance, Information Theft
ECU-IoHT	4 cat.: ARP spoofing, DoS, Smurf, PortScan
ToN-IoT	9 cat.: DoS, DDoS, Backdoor, Injection, Password, Ransomware, XSS, MITM
IoT-23	12 cat.: Okiru, Mirai, Torii, C&C-HeartBeat
CIC IoT 2023	7 cat.: DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, Mirai
CIC IoT DIAD 2024	7 cat.: DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, Mirai
CIC IoMT 2024	DDoS, DoS, Recon, MQTT, Spoofing
DataSense: CIC IIoT 2025	7 cat.: Recon, DoS, DDoS, Web, MITM, Brute Force, Malware (Mirai) + Benign; 49 attack types

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Table A.2: Distribution of anomalies in datasets.

Dataset	Normal / Benign	First largest anomaly type	Second largest anomaly type	Third largest anomaly type
UNSW-NB15	87%	Generic (8%)	Exploits (2%)	DoS (1%) Reconnaissance (1%) Fuzzers (1%)
CICIDS-2017	62%	Port Scan (12%)	DoS-Hulk (11%)	DDoS-LOIC (7%)
BoT-IoT	0%	DDoS (53%)	DoS (45%)	Reconnaissance/ Service Scan (2%)
ToN-IoT	4%	Scanning (32%)	DDoS (28%)	DoS (15%)
IoT-23	11%	Port Scan (56%)	Okiru (22%)	DDoS (11%)
CIC-IoT-DIAD	25%	Web-based (32%)	Brute Force (28%)	Mirai (3%) DoS (3%) DDoS (3%) Spoofing (3%)
DataSense: CIC IIoT Dataset 2025	< 0%	DDoS (63%)	DoS (29%)	Reconnaissance (3%) Man-in-the- Middle (7%)

Appendix B. Tables Relevant to RQ-2

Table B.1: Value-based RL algorithms used in cybersecurity.

	Alg.	Type	Action space	Policy	Family	Strengths	Limitations
C1	Q-Learning (tabular)	Model-free	Discrete	Off-policy	Value-based	Very simple and interpretable baseline; useful for small state spaces and as red-team baseline	Does not scale to high-dimensional states; weak generalization vs deep methods
	SARSA (tabular)	Model-free	Discrete	On-policy	Value-based	On-policy updates can be steadier under non-stationarity; simple to implement; common baseline	Sample-inefficient; weaker asymptotic performance; sensitive to exploration policy
	Deep SARSA	Model-free	Discrete	On-policy	Value-based	Improved multi-class intrusion detection (NSL-KDD, UNSW-NB15)	Detects only known attacks; less sample-efficient
	DQN	Model-free	Discrete	Off-policy	Value-based	Baseline for discrete actions; replay buffer improves efficiency	Overestimation bias; poor with continuous actions; unstable under distribution shift
	DDQN	Model-free	Discrete	Off-policy	Value-based	Mitigates DQN overestimation; stable with target network and replay	Discrete-only; sensitive to hyperparameters; underestimation possible
	Dueling-DQN	Model-free	Discrete	Off-policy	Value-based	Separates $V(s)$ and advantage to improve value estimates; stronger than DQN in several comparisons	Still discrete-only; benefits depend on tuning; inherits replay/target sensitivities

Table B.2: Policy-based (policy gradient) RL algorithms used in cybersecurity.

Alg.	Type	Action space	Policy	Family	Strengths	Limitations
PPO	Model-free	Continuous / Discrete	On-policy	Policy-gradient (SPG ¹)	Stable clipped updates; widely reproducible across tasks	On-policy sample cost; sensitive to clip ϵ and GAE λ
A2C	Model-free	Discrete	On-policy	Policy-gradient (SPG ¹)	Simple baseline; advantage estimation stabilizes updates	Lower performance than value-based DQN variants in ICS NIDS
A3C	Model-free	Continuous / Discrete	On-policy	Policy-gradient (SPG ¹)	Asynchronous parallel workers speed wall-clock learning; strong baseline	On-policy sample inefficiency; stale-gradient effects; tuning entropy/LR needed

Table B.3: Hybrid actor–critic RL algorithms used in cybersecurity.

Alg.	Type	Action space	Policy	Family	Strengths	Limitations
Actor–Critic (generic)	Model-free	Continuous	Varies (on/off)	Actor–Critic	Combines critic stability with expressive policy; handles large action spaces	Coupled training can diverge; tuning entropy/advantage is critical
DDPG	Model-free	Continuous	Off-policy	Actor–Critic (DPG ²)	Continuous actions for fine-grained defense; replay improves data efficiency	Exploration brittle; sensitive to reward scaling; less stable than TD3/SAC
RRIoT ³	Model-free	Continuous	Off-policy	Actor–Critic (DPG ²)	Handles sequential IoT traffic; recurrent encoder improves partial observability	Dataset imbalance; recurrent critics unstable; high compute cost
MAPPO	Model-free	Discrete	On-policy	Actor–Critic (SPG ¹)	Shared critic accelerated convergence; outperformed IPPO in maritime OT	Simulation not fully realistic; reward–cost valuation limited
SAC	Model-free	Continuous	Off-policy	Actor–Critic (Max-Entropy ⁴)	High sample efficiency and stability via entropy regularization; robust for continuous control/response	Temperature tuning sensitivity; heavier compute/replay; less used for discrete-only tasks

¹ Stochastic Policy Gradient (SPG);² Deterministic Policy Gradient (DPG);³ Recurrent Deep Deterministic Policy Gradient (RDDPG, as used in RRIoT);⁴ Soft Actor–Critic’s maximum-entropy actor–critic formulation.

Table B.4: Summary of the state-of-the-art studies based on RL algorithms for cybersecurity.

Ref.	Year	Dataset	RL algo- rithm	Task / objec- tive	Results	Strengths	Limitations
[38]	2023	ToN-IoT (GPS / Modbus / Weather)	DQN	Red teaming / at- tack planning	Accuracy = 87.8% (Weather split)	Fast convergence; strong on Weather	Lower GPS accuracy (79.7%); Modbus weaker
[11]	2024	Smart grid simulator (custom)	DDQN (CNN & GRU)	Smart grid anomaly detec- tion	Accuracy = 95.3%	Accurate minority attack de- tection; real-time adaptabil- ity	Simulation only
[39]	2023	NSL-KDD	Deep SARSA	Network intrusion detection	F1 = 84.6%	Superior minority-class detec- tion	Evaluated on older datasets
[40]	2023	Not specified	Actor-Critic	Adaptive threat detection / re- sponse	Conceptual	Supports adversarial learning; real-time policy updates	No dataset or empirical vali- dation
[41]	2023	IPMSRL (maritime OT simulation)	MAPPO & IPPO	Industrial / OT defense	Success rate = 99.5% (MAPPO)	Collaborative defense; config- urable environment	Domain-specific; complex simulator setup
[42]	2023	IoT-23	Recurrent RL (GRU & DQN)	IoT device threat detection	F1 = 80.6%	Captures sequential IoT dy- namics	High complexity; training overhead
[47]	2023	Traffic man- agement (simulated)	DRL (PPO / DDPG)	IoT-based traffic & cyber threat re- sponse	Reward +38% over baseline	Multi-agent coordination; real-time safety decisions	Environment-specific; no open dataset
[48]	2023	Enterprise network (sim- ulated)	DQN	Real-time net- work threat detection	Accuracy = 93.1%	Adaptive learning; dynamic policy updates	Limited generalizability to public benchmarks
[47]	2023	Smart city traffic (simu- lated)	DRL (policy gradient)	Smart city cy- ber-physical threat detection	Accuracy = 91.4%	Coordinated traffic-cyber re- sponse	Complex simulation; replica- tion difficult

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Table B.4: Summary of the state-of-the-art studies based on RL algorithms for cybersecurity.

Ref.	Year	Dataset	RL rithm	algo-	Task / objec- tive	Results	Strengths	Limitations
[49]	2023	CyberBattleSim	DQL (reward shaping)		Red team attack simulation & de- fense	Steps-to-goal: 27% faster convergence	Reward-optimized adaptive planning	Limited generalization beyond emulation

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Appendix C. Tables Relevant to RQ-3

Table C.1: GAN-based generative AI algorithms used in cybersecurity.

Alg.	Type	Output modality	Training paradigm	Family	Strengths	Limitations
Vanilla GAN	Generative	Network traffic / binaries	Adversarial (G vs D)	GAN	Simple baseline; effective for IDS data augmentation; learns distributional features	Training instability; mode collapse; limited scalability to high-dimensional domains
cGAN / AC-GAN	Conditional GANs	Intrusion patterns, phishing emails	Adversarial (conditional labels)	GAN	Conditional control improves targeted attack synthesis and class balancing	Sensitive to label noise; prone to overfitting; requires rich labelled data
DCGAN / WGAN	Convolutional / Wasserstein GAN	Flow features, malware images	Adversarial (improved loss)	GAN	More stable training; WGAN alleviates mode collapse; DCGAN strong for feature learning	Higher computational cost; limited temporal modelling
Adaptive GAN	Adaptive / evolving GAN	Real-time threats	Adversarial (online updates)	GAN	Adapts to changing threat vectors with minimal retraining	No validation on distributed / real networks; assumes feature stability

Table C.2: VAE-based generative AI algorithms used in cybersecurity.

Alg.	Type	Output modality	Training paradigm	Family	Strengths	Limitations
VAE	Probabilistic generative model	Traffic logs, anomaly features	Variational (ELBO)	VAE	Stable training; learns compact latent representations; useful for anomaly detection	Generates less realistic samples vs GANs; blurry / low-fidelity outputs

Table C.3: Transformer and LLM-based generative AI algorithms used in cybersecurity.

Alg.	Type	Output modality	Training paradigm	Family	Strengths	Limitations
GPT / Chat-GPT	Auto-regressive LLM	Text (logs, phishing, prompts)	Auto-regressive (self-attention)	Transformer / LLM	Strong at language-based cyber tasks (log parsing, phishing, social engineering)	Hallucinations; vulnerable to prompt injection
BERT / other transformers	Encoder-only transformer	Log data, alerts	Self-supervised	Transformer	Good for classification and embedding-based anomaly detection	Limited generative power; requires fine-tuning on domain data

Table C.4: Hybrid and emerging generative AI algorithms used in cybersecurity.

Alg.	Type	Output modality	Training paradigm	Family	Strengths	Limitations
GAN + VAE	Hybrid generative	Traffic anomalies	Combined (adversarial + variational)	Hybrid	Balances VAE stability with GAN realism; better diversity + quality	Complex training; limited benchmarks; few cyber applications yet
GAN + Transformer	Hybrid adversarial + attention	Traffic / malware / logs	Combined (adversarial + attention)	Hybrid	Captures sequential + contextual features; promising for adaptive threats	Under-explored; reproducibility issues; training overhead
Diffusion models	Probabilistic denoising generative	Synthetic attack traffic	Denoising score matching	Diffusion	High-quality synthetic data; avoids mode collapse; robust for data augmentation	Very computationally expensive; little empirical validation in cybersecurity

Table C.5: Summary of the state-of-the-art studies based on Gen AI for cybersecurity.

Ref.	Gen AI alg.	Use case / domain	Dataset	Contribution	Limitations / challenges
[9]	GANs	Intrusion detection systems	CICIDS-2017; NSL-KDD; DARPA; CTU-13; RockYou	Taxonomy of GAN-based IDS approaches; highlights cGAN, DCGAN, WGAN for realistic intrusion patterns	Relies on legacy datasets (e.g., NSL-KDD); no real-world deployment or adversarial testing
[66]	GANs	Synthetic threat data generation	Simulated datasets	Uses GANs to generate synthetic threats for IDS; supports adversarial scenario emulation for training	Not validated on high-variance attack sets; no operational deployment
[72]	GANs	Network traffic anomaly detection	Not specified	Applies GAN-based models combining feature extraction and generation	No comparison to state-of-the-art DL; lacks adversarial defense integration
[21]	GANs	Survey on GANs for IDS	CICIDS-2017; NSL-KDD	Comprehensive survey; categorizes GAN types by threat model and dataset	Uses some outdated datasets; limited coverage of newer GAN variants or multimodal threats
[73]	GANs	IDS evaluation	NSL-KDD; CICIDS-2017	Evaluates WGAN, DCGAN, ACGAN with precision/recall	Dataset limitations; no longitudinal testing; GAN stability issues under-explored
[69]	GANs	IDS enhancement	CICIDS-2017	Proposes conditional GAN for attack pattern classification	No experimental implementation; minimal GAN configuration details
[74]	GANs	Phishing email defense	None	Generates phishing emails to harden classifiers via synthetic data	Email set lacks diversity; no comparison to transformer baselines
[75]	Adaptive GAN	Real-time threat detection	None	Adaptive GAN for evolving threats with minimal retraining	Not validated in dynamic/distributed environments; assumes feature stability
[76]	GANs, VAEs	Threat simulation (malware, phishing, intrusions)	Synthetic datasets	PCA-GAN hybrid for anomaly synthesis to train robust IDS	No real-time validation; limited dataset scope; no adversarial robustness checks
[67]	GANs, transformers	Anomaly detection	CICIDS-2017	Compares GANs vs transformers on intrusion data (detection rate, convergence, scalability)	Missing transformer tuning and GAN loss curves; reproducibility unclear
[68]	GANs, Chat-GPT	Cyber threat analysis	None	Framework outlining Gen AI misuse scenarios and attack simulation potential	No implementation or experiments

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Table C.5: Summary of the state-of-the-art studies based on Gen AI for cybersecurity.

Ref.	Gen AI alg.	Use case / domain	Dataset	Contribution	Limitations / challenges
[77]	GANs, GPT	GAN vs GPT in cybersecurity	CICIDS-2017; synthetic	Contrasts GAN and GPT for anomaly detection; reports accuracy/F1	Narrow dataset scope; simple GPT models; limited external validity
[78]	LLMs	Prompt injection, jail-breaking	None	Details jailbreak techniques and risks of unrestricted LLM use	No quantitative risk model; lacks detection/mitigation methods
[71]	LLMs	Defense simulation	None	Simulation architecture for cyber training using LLM outputs	Not evaluated; no red/blue team testing
[70]	LLMs, VAEs	Adaptive threat generation	None	LLM-driven adaptive response; feedback-based defense orchestration	No SIEM/SOAR integration; latency/performance metrics absent
[79]	ChatGPT	Explainable AI in cybersecurity	None	XAI-enhanced IDS with ChatGPT explanations; user-facing interpretability	Explanations not compared to ground truth/human baseline
[80]	ChatGPT	Social / ethical risks	None	Taxonomy of Gen AI misuse and threat classes; policy/ethics overview	Lacks empirical risk evaluation; conceptual focus
[81]	ChatGPT	Explainable IDS	CICIDS-2017	Uses ChatGPT to explain IDS decisions with LLM-based rationales	No adversarial evaluation; runtime overhead not reported; reliance on closed APIs
[82]	ChatGPT, LLMs	Policy and regulation risk	None	Analysis of Gen AI risks; layered defense and audit-based mitigation framework	No implementation/validation
[10]	ChatGPT, LLMs	Automated hacking / defense	None	Conceptual analysis: AI-generated phishing, malware, social engineering	No threat modelling or mitigation strategies
[83]	ChatGPT, Bard	LLM capabilities for offence/defense	None	Compares ChatGPT vs Bard for attack vector generation (evasiveness, diversity)	Dataset unspecified; metrics mostly qualitative

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